



Utsav Gupta

B.Tech, CSE
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., CSE(1-3)	Vivekananda Institute of Professional Studies	8.95/10	2028
Senior Secondary (XII)	Banasthali Public School	88%	2024
Secondary (X)	Banasthali Public School	91%	2022

EXPERIENCE

- **Cybersecurity & Digital Forensics Intern** Dec 2025 – Feb 2026
ISEA Phase III — MeitY, Government of India | IGDTUW, Delhi
 - Forensically analyzed ~700 GB of carved data from 26 seized HDDs (NEET 2017 exam fraud case) with a custom Unix shell toolkit: executable detection, browser artifact extraction, keyword search, timeline generation.

PROJECTS

- **SAAGA — LLM Red-Teaming Framework** Independent — Sep 2026
Tools: Python, vLLM, Ollama, FastAPI, React, FAISS 
 - Automated LLM red-teaming: a planner picks attack strategies, a generator writes concise prompts, and a deterministic 4-signal ladder verifies extracted secrets (no LLM judge).
 - CLI + FastAPI/React dashboard, FAISS-backed self-improving memory; vLLM/Ollama/HuggingFace/cloud backends; evals tuned on NVIDIA DGX A100.
- **SchemaForge — Zero-Downtime DB Migration Agent** Main Contributor — Aug 2026
Tools: Python, TrueForge, MCP, Alembic, PostgreSQL 
 - Main contributor (Agent Harness Hackathon) to an AST-aware DB migration agent: a root agent fans out parallel DB/code subagents (Postgres/MCP, sandbox); deterministic impact-graph engine (tables → ORM models → attributes → endpoints).
 - Sandbox verification (Alembic, parity checks, pytest, EXPLAIN ANALYZE) behind a human gate; production apply in a single transaction with full rollback.
- **CyberChakshu — Unified SIEM with ML Attack Detection** ISEA Internship — Feb 2026
Tools: Python, TypeScript, React, PySide6, FastAPI, scikit-learn 
 - Contributed to a SIEM + forensics platform: 56+ log parsers (FTP, web, DB, firewall, SSH, mail) with regex extraction and automatic log-type detection.
 - Detection pairs a rule engine with a Random Forest classifier (25 features): MITRE ATT&CK mapping, confidence scores, and multi-log correlation into scored attack chains.
 - PySide6 desktop + React frontends (FastAPI, Hono/Cloudflare Workers); forensic disk pipeline (E01/DD → IoCs → JSON/HTML/PDF); LLM SOC analyst assistant.
- **MiniRelDB — Relational Database Engine from Scratch** Contributor — Apr 2026
Tools: Java, B+ Trees, Write-Ahead Log, LSM-tree 
 - Heavily optimized a from-scratch Java RDBMS (SQL parser, Volcano plan tree, B+ trees, WAL/LSM, 4KB pages, LRU buffer): one +3.8k/-0.3k-line diff across query, index and storage layers, plus a benchmark harness.
- **Hybrid CSPRNG (BLAKE3)** Independent — Sep 2025
Tools: Python, BLAKE3, NumPy, SciPy, Matplotlib 
 - Hybrid CSPRNG on BLAKE3 with external entropy (os.urandom, time); test suite (Chi-square, runs, autocorrelation) validated against the NIST STS battery.
 - Matplotlib distribution plots, industry-RNG benchmarks, argparse CLI for reproducible testing.

ACHIEVEMENTS & CERTIFICATIONS

- Max rating on **Codeforces**: 1014 [\[Profile\]](#)
- **NPTEL** courses: Ethical Hacking In TOP 5%, Cloud Computing [Silver Medal], Mathematical Foundation for Machine Learning, Getting Started with Competitive Programming, Discrete Mathematics.
- **Red Hat Linux**; **CHFI** certification by EC-COUNCIL; value-added course on **Computational Methods**.

SKILLS

- **Programming Languages**: Python, C, C++, Java, SQL, Bash
- **Technologies**: LLM Inference (vLLM, llama.cpp, Ollama), CUDA/HPC, Cryptography (BLAKE3, SHA, RSA, RNG design), DB Internals (query planning, B+ trees, WAL), SIEM, Digital Forensics, SDL2
- **Tools**: Git, NumPy, SciPy, Matplotlib, scikit-learn, FAISS, PySide6, BurpSuite, Wireshark, NMAP, Autopsy, Sleuth Kit, OpenSSL, Hashcat, MATLAB, OpenBullet, SilverBullet